Fundamentals of the LLM by Andrej Karpehyra

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=> Deep Dive into llms :—

1. Step :- Pretraining :—

The FineWeb recipe → hugging Face

In the next subsections we will explain each of the steps taken to produce the FineWeb dataset.

You can find a fully reproducible datatrove config here.



The FineWeb pipeline

=> Text → binary (2^8) bit → Octal

$(01101011)_2$ $(153)_8$

=> Common numbers

=> Text → Tokenization

↳ case - sensitive

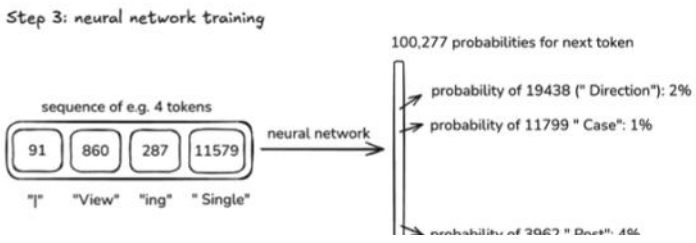
↳ for visualization :— (TikTokenizer)

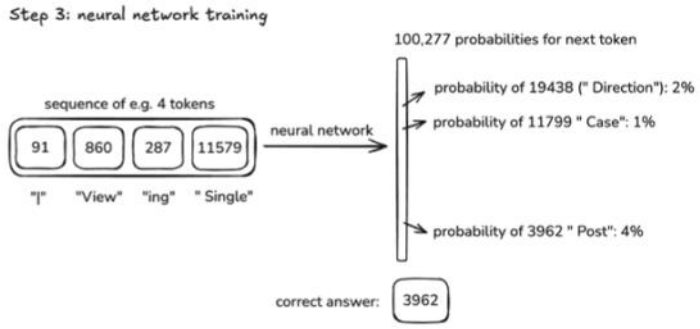
=> Pretraining :—

Step 1 :- download and preprocess the internet (The Fine web pipeline)

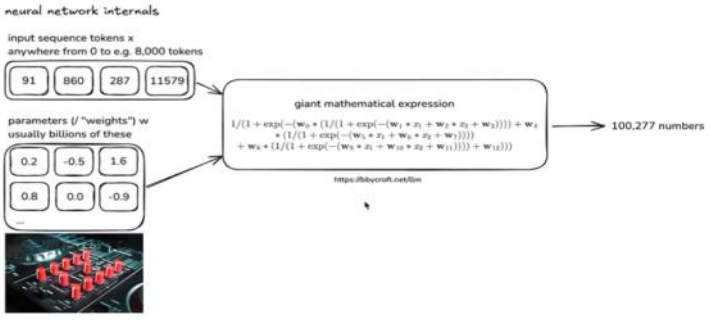
Step 2 :- Tokenization :-

Step 3 :- neural network training



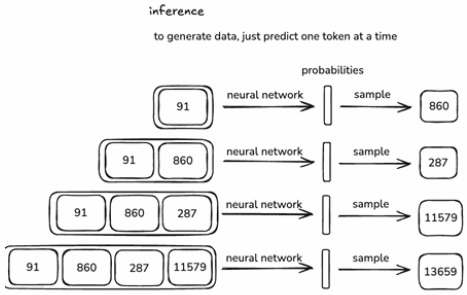


=>



=> Step-4 :-

=> Inference :-



* Inference means feeding tokens to generate next token.

=> Demo :-

Demo: reproducing OpenAI's GPT-2

GPT-2 was published by OpenAI in 2019
Paper: "Language Models are Unsupervised Multitask Learners"

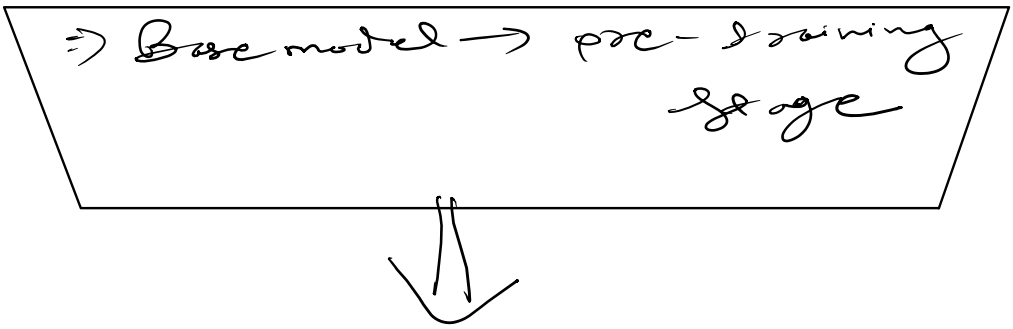
Transformer neural network with:
- 1.6 billion parameters
- maximum context length of 1024 tokens
- trained on about 100 billion tokens

My reproduction with llm.c:
https://github.com/karpathy/llm.c/discussions/677

=> base model → (1) gpt-2

(2) Llama-3

→ for visualization (hyperbolic website)



⇒ Compact Summary :—

(1) LLM → large language model

⇒ Today → pre training of the llm

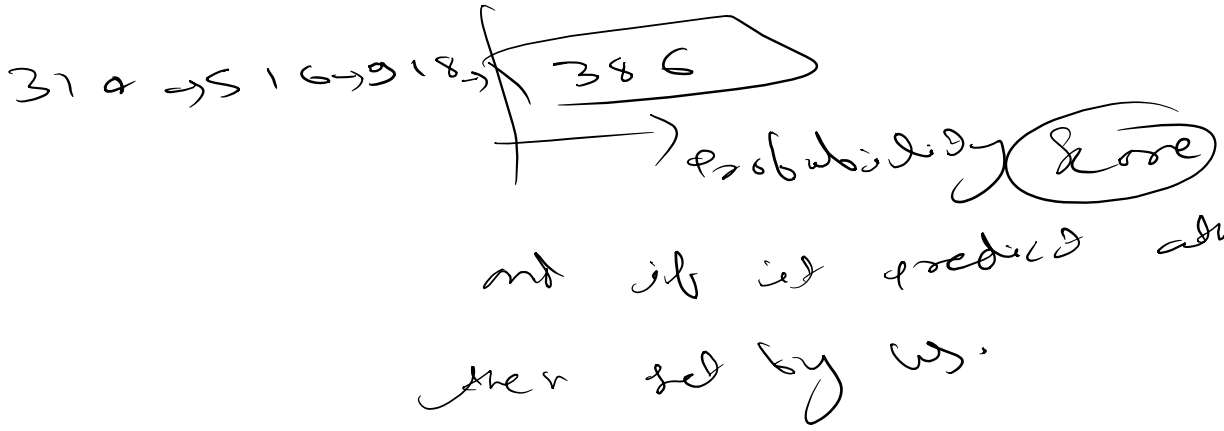
(1) gathering data from the entire internet
↳ Hugging face (Fineweb)
↳

(2) Data filter → remove confidential information and check the confidence score of the language > 0.65 then ready for the train

(3) Convert into the binary or octal numbers and then common

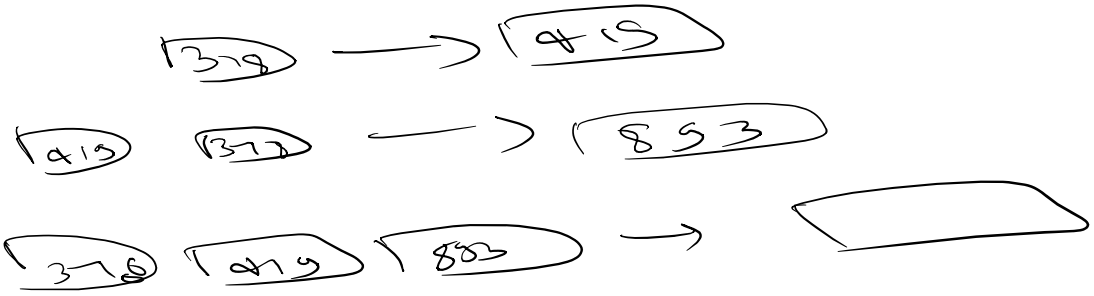
Octal numbers and then Common numbers recognized are numbers and these are called tokens :-

=> LLMs → next token prediction



not it is predicted after then fed by us.

=> Intersone :-



Base model

→ It is simple prediction model.